
-- APB DIRECTED VERIFICATION TEST SUITE --
-- Self-Checking | Directed | Icarus Compatible --

Clock Period: 10 ns

Timeout: 20 cycles

Read Pipeline Delay: 2 cycles

[0] INFO: Applying Reset

[70000] INFO: Reset Complete

TEST GROUP 1: Basic Functionality

[110000] INFO: WRITE 0x10=0xaa
[135000] WARNING: Transfer request during active transaction
[145000] INFO: WRITE PASS: 0x10 = 0xaa
[270000] INFO: READ 0x10
[315000] INFO: READ PASS: 0x10 = 0xaa (pipeline check)
[410000] INFO: WRITE-READ 0x20
[430000] INFO: WRITE 0x20=0x55
[465000] INFO: WRITE PASS: 0x20 = 0x55
[590000] INFO: READ 0x20
[635000] INFO: READ PASS: 0x20 = 0x55 (pipeline check)
[750000] INFO: WRITE 0x30=0xaa
[785000] INFO: WRITE PASS: 0x30 = 0xaa
[910000] INFO: WRITE 0x31=0xab
[945000] INFO: WRITE PASS: 0x31 = 0xab
[1070000] INFO: WRITE 0x32=0xac
[1105000] INFO: WRITE PASS: 0x32 = 0xac
[1230000] INFO: WRITE 0x33=0xad
[1265000] INFO: WRITE PASS: 0x33 = 0xad
[1390000] INFO: READ 0x30
[1435000] INFO: READ PASS: 0x30 = 0xaa (pipeline check)
[1550000] INFO: READ 0x31
[1595000] INFO: READ PASS: 0x31 = 0xab (pipeline check)
[1710000] INFO: READ 0x32
[1755000] INFO: READ PASS: 0x32 = 0xac (pipeline check)
[1870000] INFO: READ 0x33
[1915000] INFO: READ PASS: 0x33 = 0xad (pipeline check)

TEST GROUP 2: Error & Boundary Conditions

[2030000] INFO: WRITE 0x80=0x55
[2065000] INFO: ERROR PASS: 0x80 correctly invalid
[2190000] INFO: READ 0x80
[2225000] INFO: ERROR PASS: 0x80 correctly invalid
[2330000] INFO: WRITE-READ 0x7f
[2350000] INFO: WRITE 0x7f=0xff
[2385000] INFO: WRITE PASS: 0x7f = 0xff
[2510000] INFO: READ 0x7f
[2555000] INFO: READ PASS: 0x7f = 0xff (pipeline check)
[2670000] INFO: WRITE 0xff=0x0
[2705000] INFO: ERROR PASS: 0xff correctly invalid
[2810000] INFO: WRITE-READ 0x0
[2830000] INFO: WRITE 0x0=0x12
[2865000] INFO: WRITE PASS: 0x0 = 0x12
[2990000] INFO: READ 0x0
[3035000] INFO: READ PASS: 0x0 = 0x12 (pipeline check)

TEST GROUP 3: Stress & Corner Cases

[3130000] INFO: BURST WRITE 8 addresses from 0x40
[3150000] INFO: WRITE 0x40=0xa0
[3185000] INFO: WRITE PASS: 0x40 = 0xa0
[3330000] INFO: WRITE 0x41=0xa1
[3365000] INFO: WRITE PASS: 0x41 = 0xa1
[3510000] INFO: WRITE 0x42=0xa2
[3545000] INFO: WRITE PASS: 0x42 = 0xa2
[3690000] INFO: WRITE 0x43=0xa3
[3725000] INFO: WRITE PASS: 0x43 = 0xa3
[3870000] INFO: WRITE 0x44=0xa4
[3905000] INFO: WRITE PASS: 0x44 = 0xa4
[4050000] INFO: WRITE 0x45=0xa5
[4085000] INFO: WRITE PASS: 0x45 = 0xa5
[4230000] INFO: WRITE 0x46=0xa6
[4265000] INFO: WRITE PASS: 0x46 = 0xa6
[4410000] INFO: WRITE 0x47=0xa7
[4445000] INFO: WRITE PASS: 0x47 = 0xa7
[4570000] INFO: BURST READ 8 addresses from 0x40
[4590000] INFO: READ 0x40
[4635000] INFO: READ PASS: 0x40 = 0xa0 (pipeline check)
[4770000] INFO: READ 0x41
[4815000] INFO: READ PASS: 0x41 = 0xa1 (pipeline check)
[4950000] INFO: READ 0x42
[4995000] INFO: READ PASS: 0x42 = 0xa2 (pipeline check)
[5130000] INFO: READ 0x43
[5175000] INFO: READ PASS: 0x43 = 0xa3 (pipeline check)
[5310000] INFO: READ 0x44
[5355000] INFO: READ PASS: 0x44 = 0xa4 (pipeline check)
[5490000] INFO: READ 0x45
[5535000] INFO: READ PASS: 0x45 = 0xa5 (pipeline check)
[5670000] INFO: READ 0x46
[5715000] INFO: READ PASS: 0x46 = 0xa6 (pipeline check)
[5850000] INFO: READ 0x47
[5895000] INFO: READ PASS: 0x47 = 0xa7 (pipeline check)
[6010000] INFO: WRITE-READ 0x50
[6030000] INFO: WRITE 0x50=0xff
[6065000] INFO: WRITE PASS: 0x50 = 0xff
[6190000] INFO: READ 0x50
[6235000] INFO: READ PASS: 0x50 = 0xff (pipeline check)
[6330000] INFO: WRITE-READ 0x51
[6350000] INFO: WRITE 0x51=0x0
[6385000] INFO: WRITE PASS: 0x51 = 0x0
[6510000] INFO: READ 0x51
[6555000] INFO: READ PASS: 0x51 = 0x0 (pipeline check)
[6670000] INFO: WRITE 0x60=0xb0
[6705000] INFO: WRITE PASS: 0x60 = 0xb0

[6830000] INFO: READ 0x60
[6875000] INFO: READ PASS: 0x60 = 0xb0 (pipeline check)
[6990000] INFO: WRITE 0x61=0xb1
[7025000] INFO: WRITE PASS: 0x61 = 0xb1
[7150000] INFO: READ 0x61
[7195000] INFO: READ PASS: 0x61 = 0xb1 (pipeline check)
[7310000] INFO: WRITE 0x62=0xb2
[7345000] INFO: WRITE PASS: 0x62 = 0xb2
[7470000] INFO: READ 0x62
[7515000] INFO: READ PASS: 0x62 = 0xb2 (pipeline check)
[7630000] INFO: WRITE 0x63=0xb3
[7665000] INFO: WRITE PASS: 0x63 = 0xb3
[7790000] INFO: READ 0x63
[7835000] INFO: READ PASS: 0x63 = 0xb3 (pipeline check)
[7950000] INFO: WRITE 0x64=0xb4
[7985000] INFO: WRITE PASS: 0x64 = 0xb4
[8110000] INFO: READ 0x64
[8155000] INFO: READ PASS: 0x64 = 0xb4 (pipeline check)

TEST GROUP 4: Reset Recovery

[8250000] INFO: Starting: Reset During Transaction
[8325000] INFO: WRITE PASS: 0x70 = 0xaa
[8340000] INFO: Reset During Transaction Complete
[8410000] INFO: Starting: Reset then Transaction
[8500000] INFO: WRITE-READ 0x75
[8520000] INFO: WRITE 0x75=0xbb
[8555000] INFO: WRITE PASS: 0x75 = 0xbb
[8680000] INFO: READ 0x75
[8725000] INFO: READ PASS: 0x75 = 0xbb (pipeline check)
[8820000] INFO: Reset Cycle 1
[8910000] INFO: WRITE-READ 0x78
[8930000] INFO: WRITE 0x78=0xcc
[8965000] INFO: WRITE PASS: 0x78 = 0xcc
[9090000] INFO: READ 0x78
[9135000] INFO: READ PASS: 0x78 = 0xcc (pipeline check)
[9230000] INFO: Reset Cycle 2
[9320000] INFO: WRITE-READ 0x79
[9340000] INFO: WRITE 0x79=0xcd
[9375000] INFO: WRITE PASS: 0x79 = 0xcd
[9500000] INFO: READ 0x79
[9545000] INFO: READ PASS: 0x79 = 0xcd (pipeline check)
[9640000] INFO: Reset Cycle 3
[9730000] INFO: WRITE-READ 0x7a
[9750000] INFO: WRITE 0x7a=0xce
[9785000] INFO: WRITE PASS: 0x7a = 0xce
[9910000] INFO: READ 0x7a
[9955000] INFO: READ PASS: 0x7a = 0xce (pipeline check)

TEST GROUP 5: Performance Measurement

[10050000] INFO: Starting: Performance Measurement
[10070000] INFO: WRITE 0x7a=0xdd
[10105000] INFO: WRITE PASS: 0x7a = 0xdd
[10240000] INFO: WRITE 0x7b=0xde
[10275000] INFO: WRITE PASS: 0x7b = 0xde
[10410000] INFO: WRITE 0x7c=0xdf
[10445000] INFO: WRITE PASS: 0x7c = 0xdf
[10580000] INFO: WRITE 0x7d=0xe0
[10615000] INFO: WRITE PASS: 0x7d = 0xe0
[10750000] INFO: WRITE 0x7e=0xe1
[10785000] INFO: WRITE PASS: 0x7e = 0xe1
[10920000] INFO: WRITE 0x7a=0xe2
[10955000] INFO: WRITE PASS: 0x7a = 0xe2
[11090000] INFO: WRITE 0x7b=0xe3
[11125000] INFO: WRITE PASS: 0x7b = 0xe3
[11260000] INFO: WRITE 0x7c=0xe4
[11295000] INFO: WRITE PASS: 0x7c = 0xe4
[11430000] INFO: WRITE 0x7d=0xe5
[11465000] INFO: WRITE PASS: 0x7d = 0xe5
[11600000] INFO: WRITE 0x7e=0xe6
[11635000] INFO: WRITE PASS: 0x7e = 0xe6
[11770000] INFO: WRITE 0x7a=0xe7
[11805000] INFO: WRITE PASS: 0x7a = 0xe7
[11940000] INFO: WRITE 0x7b=0xe8
[11975000] INFO: WRITE PASS: 0x7b = 0xe8
[12110000] INFO: WRITE 0x7c=0xe9
[12145000] INFO: WRITE PASS: 0x7c = 0xe9
[12280000] INFO: WRITE 0x7d=0xea
[12315000] INFO: WRITE PASS: 0x7d = 0xea
[12450000] INFO: WRITE 0x7e=0xeb
[12485000] INFO: WRITE PASS: 0x7e = 0xeb
[12620000] INFO: WRITE 0x7a=0xec
[12655000] INFO: WRITE PASS: 0x7a = 0xec
[12790000] INFO: WRITE 0x7b=0xed
[12825000] INFO: WRITE PASS: 0x7b = 0xed
[12960000] INFO: WRITE 0x7c=0xee
[12995000] INFO: WRITE PASS: 0x7c = 0xee
[13130000] INFO: WRITE 0x7d=0xef
[13165000] INFO: WRITE PASS: 0x7d = 0xef
[13300000] INFO: WRITE 0x7e=0xf0
[13335000] INFO: WRITE PASS: 0x7e = 0xf0
[13470000] INFO: WRITE 0x7a=0xf1
[13505000] INFO: WRITE PASS: 0x7a = 0xf1
[13640000] INFO: WRITE 0x7b=0xf2
[13675000] INFO: WRITE PASS: 0x7b = 0xf2
[13810000] INFO: WRITE 0x7c=0xf3
[13845000] INFO: WRITE PASS: 0x7c = 0xf3

[13980000] INFO: WRITE 0x7d=0xf4
[14015000] INFO: WRITE PASS: 0x7d = 0xf4
[14150000] INFO: WRITE 0x7e=0xf5
[14185000] INFO: WRITE PASS: 0x7e = 0xf5
[14320000] INFO: WRITE 0x7a=0xf6
[14355000] INFO: WRITE PASS: 0x7a = 0xf6
[14490000] INFO: WRITE 0x7b=0xf7
[14525000] INFO: WRITE PASS: 0x7b = 0xf7
[14660000] INFO: WRITE 0x7c=0xf8
[14695000] INFO: WRITE PASS: 0x7c = 0xf8
[14830000] INFO: WRITE 0x7d=0xf9
[14865000] INFO: WRITE PASS: 0x7d = 0xf9
[15000000] INFO: WRITE 0x7e=0xfa
[15035000] INFO: WRITE PASS: 0x7e = 0xfa
[15170000] INFO: WRITE 0x7a=0xfb
[15205000] INFO: WRITE PASS: 0x7a = 0xfb
[15340000] INFO: WRITE 0x7b=0xfc
[15375000] INFO: WRITE PASS: 0x7b = 0xfc
[15510000] INFO: WRITE 0x7c=0xfd
[15545000] INFO: WRITE PASS: 0x7c = 0xfd
[15680000] INFO: WRITE 0x7d=0xfe
[15715000] INFO: WRITE PASS: 0x7d = 0xfe
[15850000] INFO: WRITE 0x7e=0xff
[15885000] INFO: WRITE PASS: 0x7e = 0xff
[16020000] INFO: WRITE 0x7a=0x0
[16055000] INFO: WRITE PASS: 0x7a = 0x0
[16190000] INFO: WRITE 0x7b=0x1
[16225000] INFO: WRITE PASS: 0x7b = 0x1
[16360000] INFO: WRITE 0x7c=0x2
[16395000] INFO: WRITE PASS: 0x7c = 0x2
[16530000] INFO: WRITE 0x7d=0x3
[16565000] INFO: WRITE PASS: 0x7d = 0x3
[16700000] INFO: WRITE 0x7e=0x4
[16735000] INFO: WRITE PASS: 0x7e = 0x4
[16870000] INFO: WRITE 0x7a=0x5
[16905000] INFO: WRITE PASS: 0x7a = 0x5
[17040000] INFO: WRITE 0x7b=0x6
[17075000] INFO: WRITE PASS: 0x7b = 0x6
[17210000] INFO: WRITE 0x7c=0x7
[17245000] INFO: WRITE PASS: 0x7c = 0x7
[17380000] INFO: WRITE 0x7d=0x8
[17415000] INFO: WRITE PASS: 0x7d = 0x8
[17550000] INFO: WRITE 0x7e=0x9
[17585000] INFO: WRITE PASS: 0x7e = 0x9
[17720000] INFO: WRITE 0x7a=0xa
[17755000] INFO: WRITE PASS: 0x7a = 0xa
[17890000] INFO: WRITE 0x7b=0xb
[17925000] INFO: WRITE PASS: 0x7b = 0xb
[18060000] INFO: WRITE 0x7c=0xc
[18095000] INFO: WRITE PASS: 0x7c = 0xc
[18230000] INFO: WRITE 0x7d=0xd
[18265000] INFO: WRITE PASS: 0x7d = 0xd
[18400000] INFO: WRITE 0x7e=0xe

[18435000] INFO: WRITE PASS: 0x7e = 0xe
Performance: 50 writes in 8500 ns
Average: 170 ns per write

FINAL TEST SUMMARY		
Total Tests :	108	
Passed :	108	
Failed :	0	
Warnings :	1	
Pass Rate :	100.0%	

? [PASS] ALL TESTS PASSED
? ?? 1 WARNINGS - Review Log

\$finish called from file "design.sv", line 1177.
\$finish at simulation time 18670000
VCS Simulation Report
Time: 18670000 ps
CPU Time: 0.590 seconds; Data structure size: 0.0Mb
Thu Oct 1 16:52:53 2026
Finding VCD file...
./apb_sim.vcd
[2026-10-01 20:52:53 UTC] Opening EPWave...
Done